

Inferential Statistics: Knowledge Score vs. Education Level

One-Way ANOVA (Analysis of Variance)

This report examines whether there is a statistically significant difference in clinical knowledge across four different educational attainment levels among thalassemia carriers.

Descriptive Statistics

Education Level	N	Mean (V3 Score)	Std Dev (V3 Score)	Mean (Weighted V3)	Std Dev (Weighted)
Up to O/L	37	6.62	7.04	-0.649	3.727
Up to A/L	83	9.24	8.57	0.715	4.574
Undergraduate	43	16.51	5.29	5.032	3.484
Graduate	36	14.72	6.56	3.885	3.840

Hypothesis Testing (One-Way ANOVA)

We use an Analysis of Variance (ANOVA) to determine if the means of the four independent groups are significantly different from each other.

1. Analysis using the mathematically rigorous Weighted V3 Knowledge Score:

- **F-Statistic:** 18.439
- **P-Value:** 0.0000
- **Conclusion: Statistically Significant ($p < 0.05$).** There is a meaningful difference in knowledge based on education level.

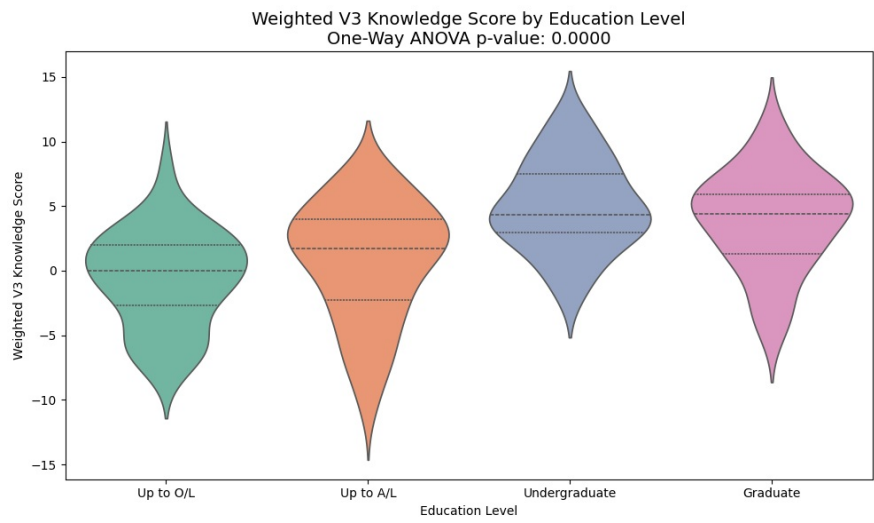
2. Analysis using the standard V3 Knowledge Score:

- **F-Statistic:** 17.041
- **P-Value:** 0.0000
- **Conclusion: Statistically Significant ($p < 0.05$).**

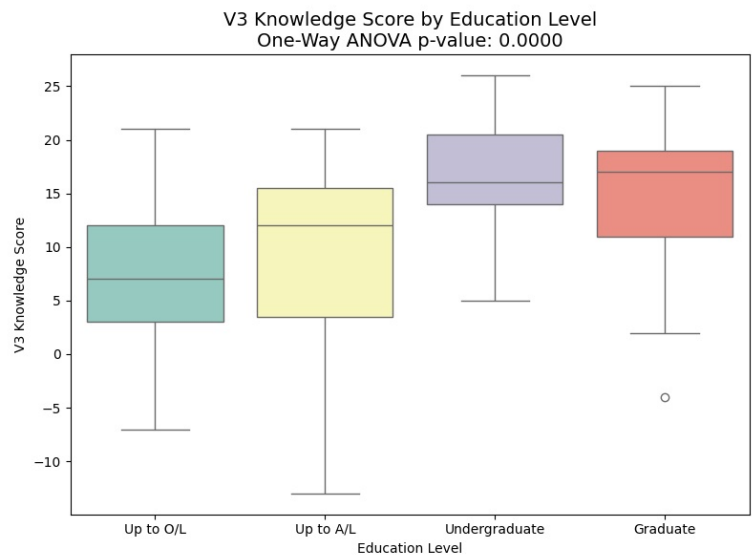
Visualizations

Weighted V3 Score Violin Plot

The violin plot demonstrates the distribution density for each educational group.



Standard V3 Score Box Plot



Box Plot